

Supplementary Information for

Undoped Ruthenium Oxide as a Stable Catalyst for the Acidic Oxygen Evolution Reaction

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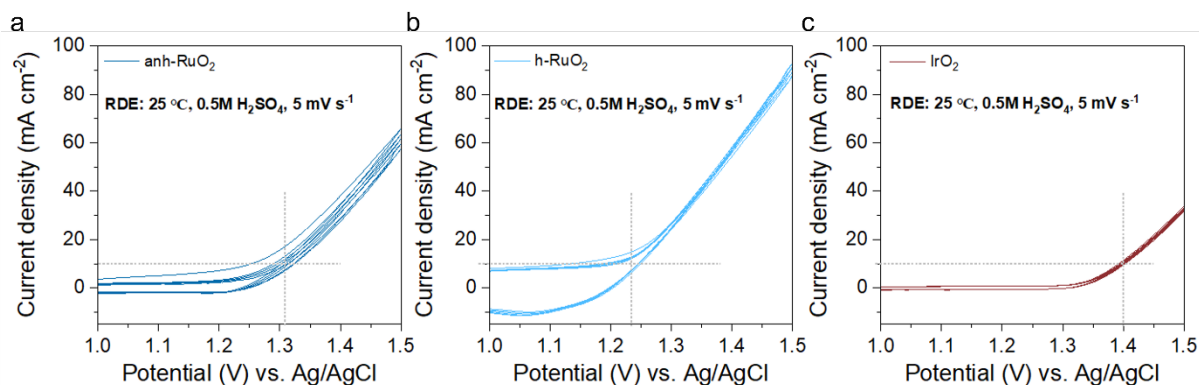


Figure S1. Cyclic voltammetry curves of the anh-RuO₂, h-RuO_x, and commercial IrO₂ catalysts obtained under RDE system.

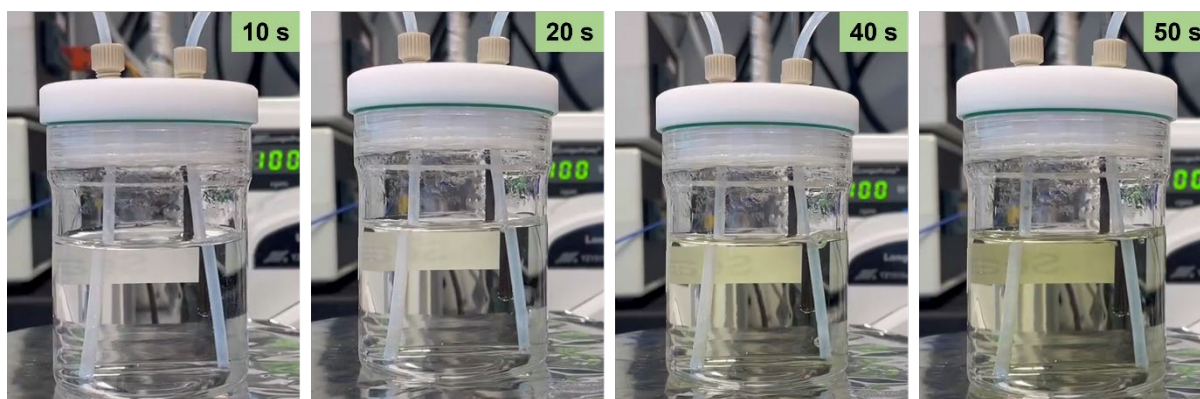


Figure S2. Observation of anode circulating water changes during the PEMWE operation with the h-RuO_x anode.

Table S1. Ru concentration in the anode circulating water detected by ICP-MS during the stability test of the PEMWEs with different Ru oxide anodes.

Sample	Concentration (ppm)
An: anh-RuO ₂ 1 A cm ⁻² after 100 h	0.002
An: anh-RuO ₂ 1 A cm ⁻² after 200 h	0.003
An: anh-RuO ₂ 1 A cm ⁻² after 300 h	0.002
An: anh-RuO ₂ 1 A cm ⁻² after 400 h	0.003
An: anh-RuO ₂ 1 A cm ⁻² after 500 h	0.004
An: h-RuO _x 1 A cm ⁻² after failure	0.007

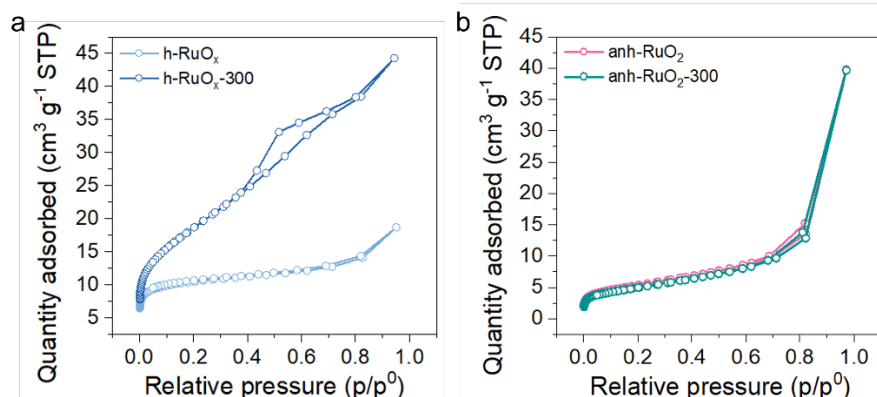


Figure S3. N₂ adsorption-desorption isotherms of **a**, the h-RuO_x before and after the thermal treatment conducted under 300 °C for 2 h with the calculated BET surface area of 34.4 m² g⁻¹ (before) and 66.6 m² g⁻¹ (after), and **b**, the anh-RuO₂ with the calculated BET surface area of 18.8 m² g⁻¹ (before 300 °C treatment) and 17.6 m² g⁻¹ (after 300 °C treatment) respectively.

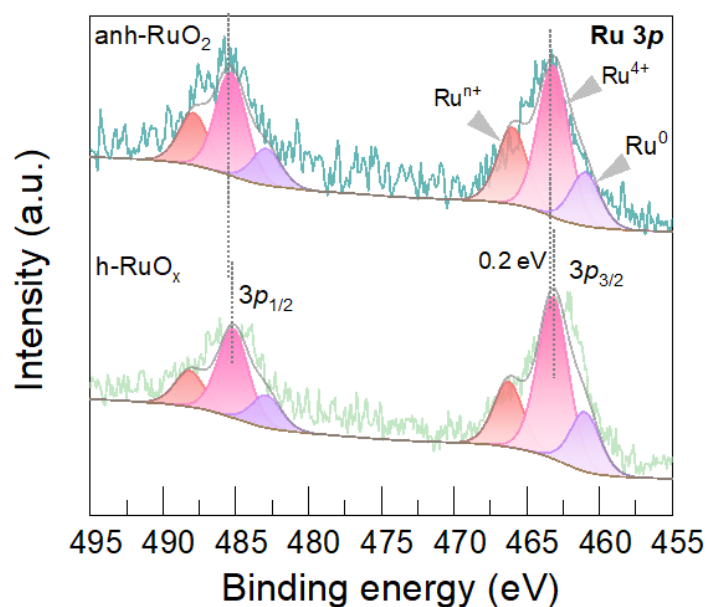


Figure S4. High-resolution Ru 3p spectra and the deconvoluted peaks.

Table S2. Development of Ru oxides and doped-Ru-oxide catalysts with improved stability for PEMWEs.

Materials	Property (surface area, and/or crystal structure)	Synthesis temperature (°C)	Stability and test conditions	Ref.
m-RuO ₂	Ru-Ru interaction formed with the	Anneal in air at 480 °C with urea	1 A cm ⁻² for 100 h with decay	1

	existence of oxygen vacancy		(ultrapure water, 60 °C)	
r-RuO ₂	calculated particle size of 15.0 nm, crystal lattice	Anneal in air at 480 °C without urea	1 A cm ⁻² for less than 20 h and fail (ultrapure water, 60 °C)	1
GB-RuO ₂	crystal structure with lattice stretching	Anneal at 300 °C in O ₂ for 2 h	100 mA cm ⁻² stabilize for 100 h (deionized water, 80 °C)	2
In-RuO ₂ /G	high crystallinity of RuO ₂ ; the incorporation of In (11 wt.%) bring oxygen vacancies	Anneal at 300 °C for 3 h	100 mA cm ⁻² for 350 h with decay (0.5 M H ₂ SO ₄ , 50 °C)	3
Pd _{0.08} Ru _{0.92} O ₂	rutile-phase RuO ₂	Anneal under air at 400–550 °C	200 mA cm ⁻² for 130 h (deionized water, 80 °C)	4
Rh-RuO ₂	1.38 wt.% of Rh doping, rutile-structured RuO ₂	Anneal under air at 300 °C for 3 h	50 mA cm ⁻² for 700 h on Ti mesh (0.5 M H ₂ SO ₄)	5
SnRuO _x	Sn:Ru ratio of around 1:1	Precursors in organic solution, then anneal under 350 °C in air for 5 h	1 A cm ⁻² for 1300 h (deionized water, 50 °C)	6
Porous and defective RuO ₂ nanosheets	amorphous Ru(OH) _x	350 °C in air for 2 h	1 A cm ⁻² for 10 h (deionized water, 90 °C)	7
Ni-RuO ₂	1.4 at.% of Ni doping	900 °C in Ar/H ₂ for 2 h first, then air-annealing at 450 °C for 3 h	200 mA cm ⁻² for 1000 h (0.1 M HClO ₄ , Room temperature)	8
Nd _{0.1} RuO _x /CC	rutile-phase RuO ₂	600 °C in air for 180 h	10 mA cm ⁻² for 50 h (0.5 M H ₂ SO ₄)	9
Pt-doped RuO ₂ hollow nanospheres	Ru:Pt ratio of 23.0:1.1	Hydrothermal at 180 °C for 12 hours, then anneal under 300°C in air for 10 hours	100 mA cm ⁻² for 100 h with visible decrease (pure water)	10
PtCo-RuO ₂	rutile-phase RuO ₂	Hydrothermal at 280 °C for 2 hours, then anneal under 400°C in O ₂ /N ₂ for 1 hours	1 A cm ⁻² for 24 h (distilled water, 80 °C)	11
Ru ₃ PbO _x	high crystallinity, Ru:Pb atomic ratio of 3:1	Anneal under Ar/H ₂ at 1000 °C for 2 h, then at 350 °C in air for 3 h	100 mA cm ⁻² for 300 h (deionized water, Room temperature)	12
W _{0.2} Er _{0.1} Ru _{0.7} O _{2-δ}	/	Anneal under 300 °C for 4h, then 400 °C for 2 h.	100 mA cm ⁻² for 120 h (0.5 M H ₂ SO ₄)	13

*The stability results presented in this table include two electrode systems mimicking PEMWE operating conditions but are not assembled with a zero-gap configuration.

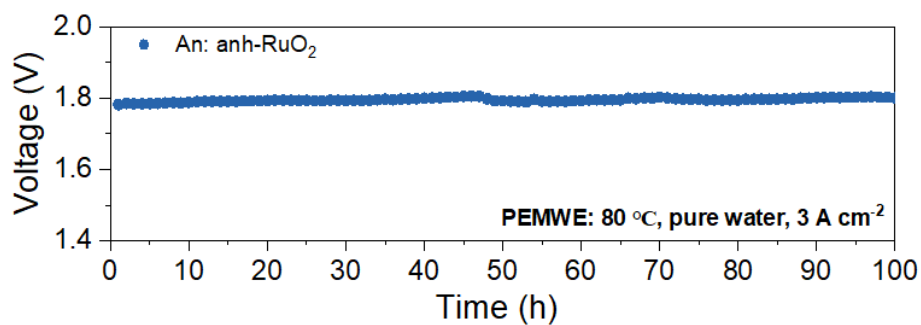


Figure S5. An operation of PEMWE with the anh-RuO₂ anode at 3 A cm⁻² for 100 h, then the anode Ru electronic structures were characterized and as shown in Figure 3.

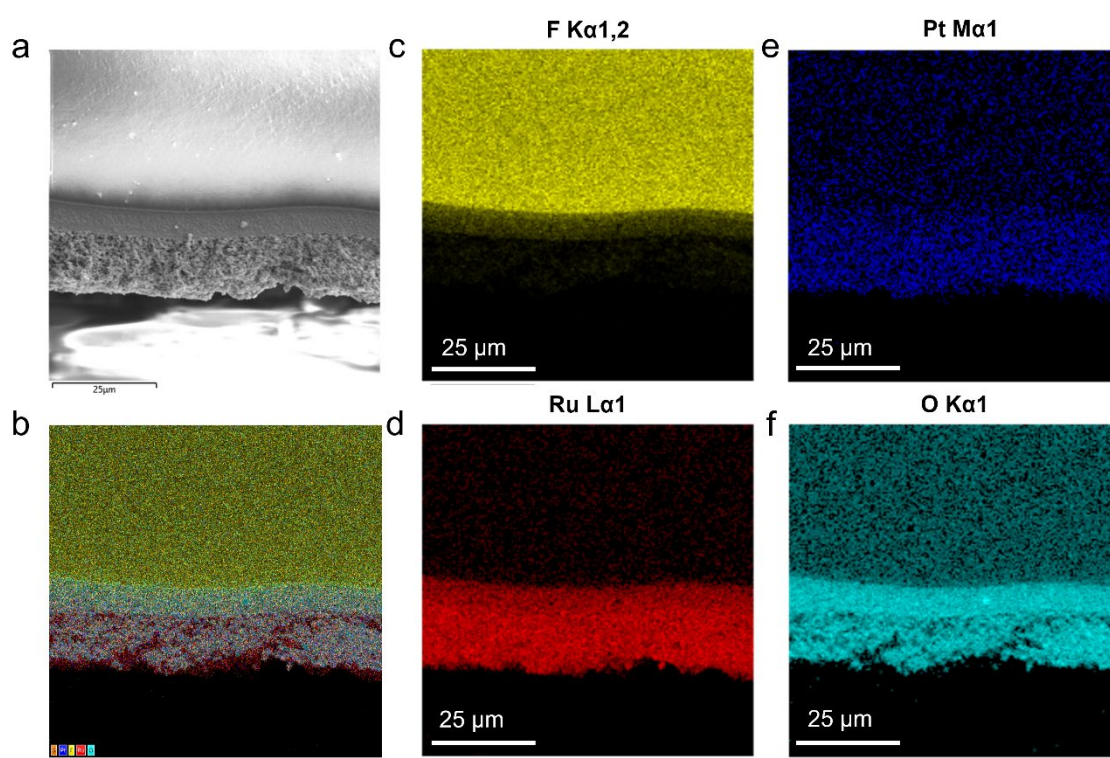


Figure S6. **a**, The anh-RuO₂ anode structure after the stability test for 550 h and **b-f**, the EDS elemental mapping of the Pt, Ru, O and F distribution showing the delamination occurred in the anode.

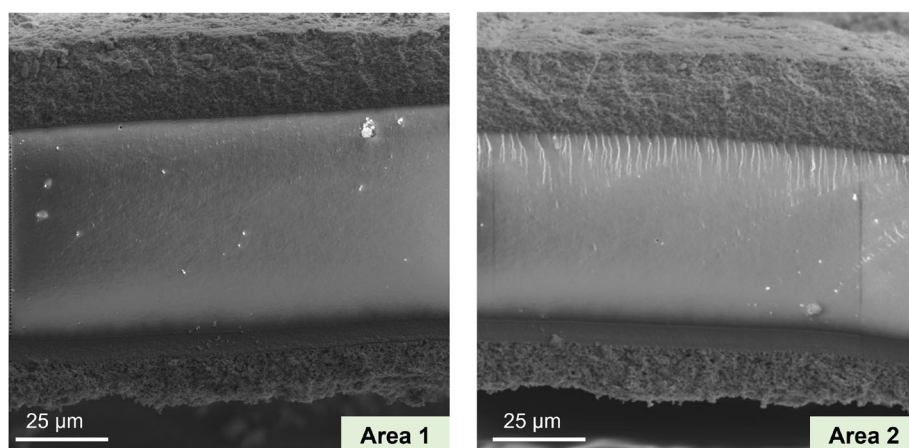


Figure S7. SEM images of the CCM with the an-RuO₂ anode after 550-h PEMWE stability test under 1 A cm⁻².

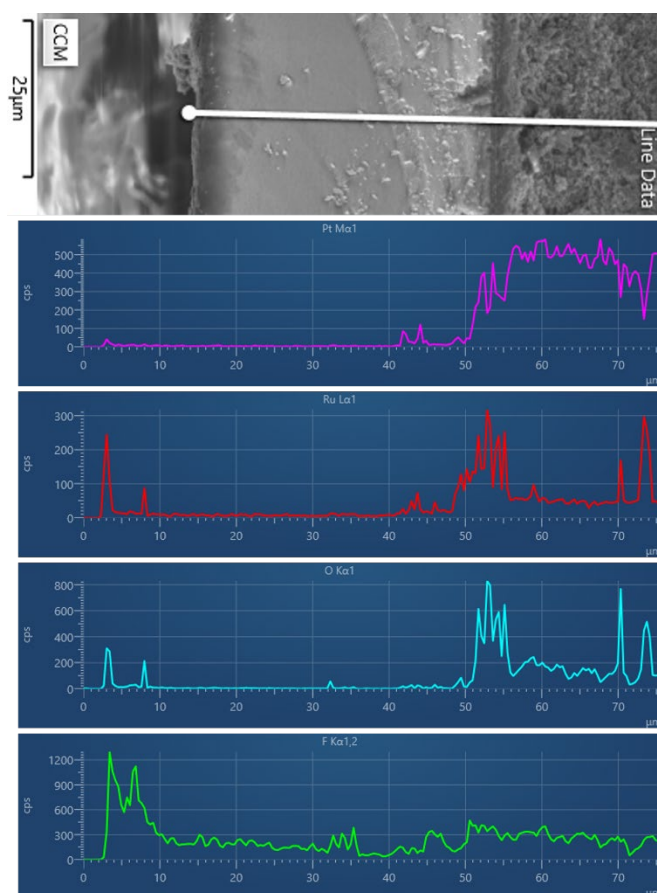


Figure S8. EDS line scanning through the CCM with the h-RuO_x anode after the failure in PEMWE stability test and the elemental distribution of Pt, Ru, O and F.

Table S3. Calculated crystallinity and crystal size of the h-RuO_x and the anh-RuO₂ after thermal treatments.

Powder	h-RuO _x -300	h-RuO _x -400	h-RuO _x -600	h-RuO _x -800	anh-RuO ₂ -300	anh-RuO ₂ -400	anh-RuO ₂ -600	anh-RuO ₂ -800
Crystallinity (%)	40.2	75	83.5	88	88.5	89.3	89.6	92.6
Crystallite size (nm)	3.9	11.8	23	46.9	38.4	39.9	60	108.8

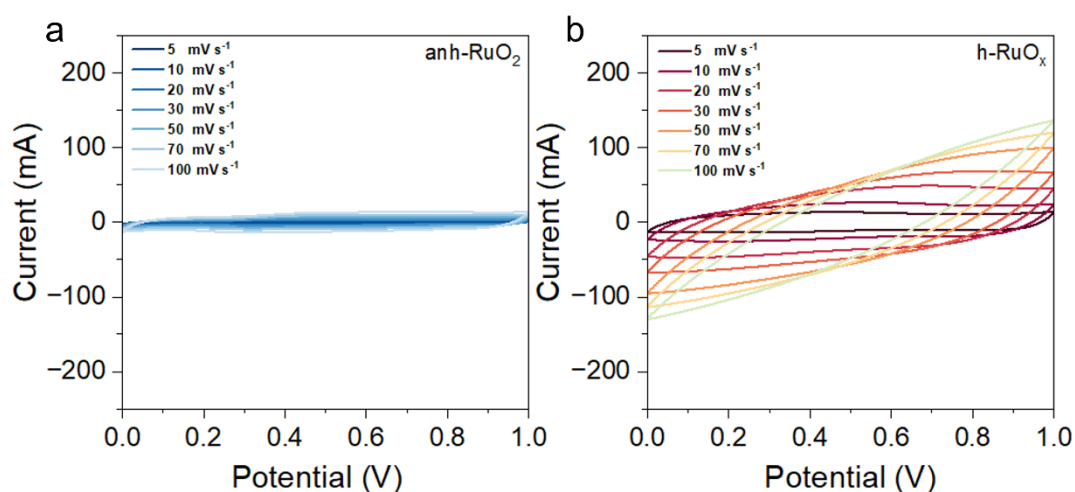


Figure S9. CV curves obtained at various scan rates from 5 to 100 mV s⁻¹ in 1M H₂SO₄ showing the proton insertion capability of **a**, the anh-RuO₂ and **b**, the h-RuO_x.

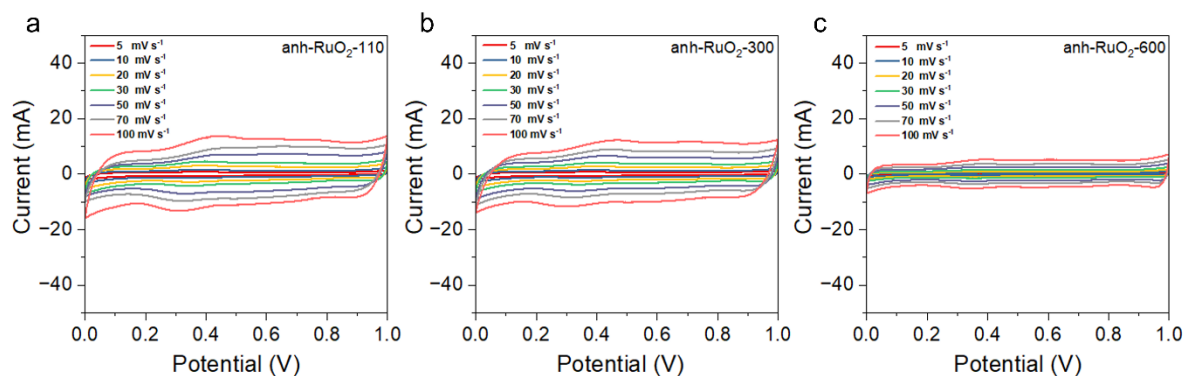


Figure S10. CV curves at various scan rates of the anh-RuO₂ after the thermal treatment conducted under 110, 300, and 600 °C respectively.

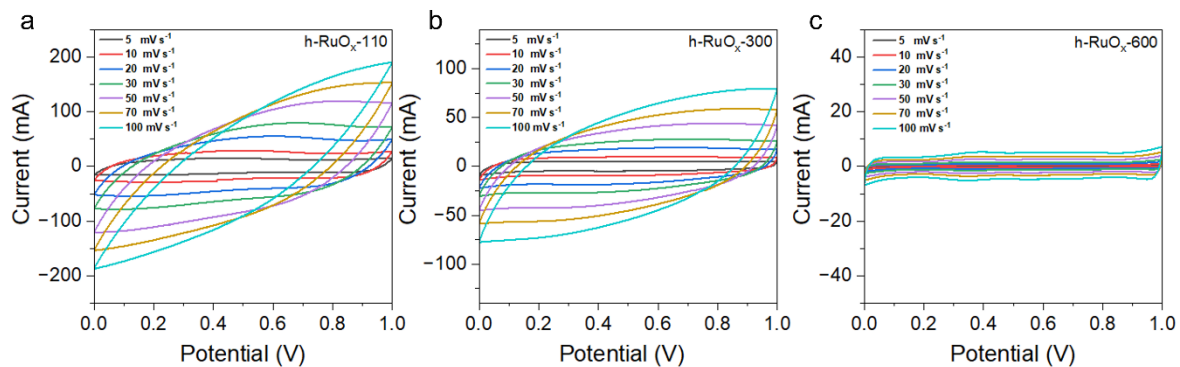


Figure S11. CV curves at various scan rates of the h-RuO_x after the thermal treatment conducted under 110, 300, and 600 °C respectively.

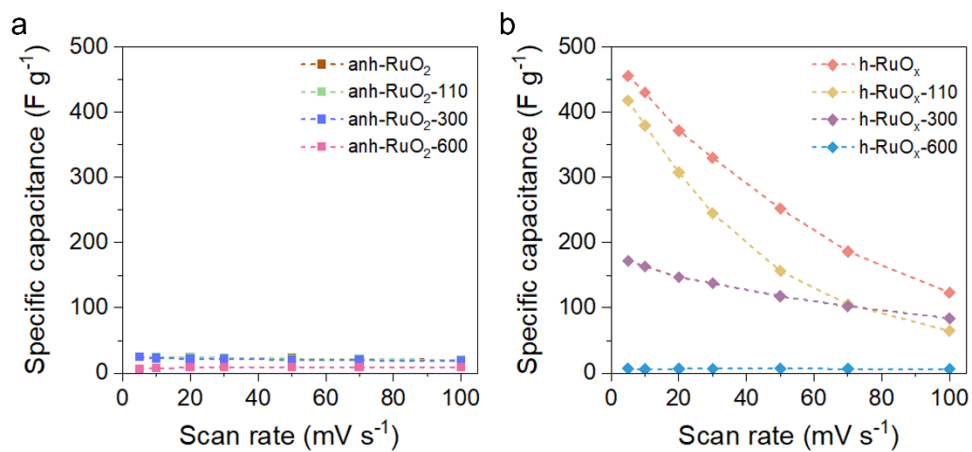


Figure S12. Specific capacitance as a function of scan rates of **a**, the anh-RuO₂ and **b**, the h-RuO_x before and after the thermal treatments.

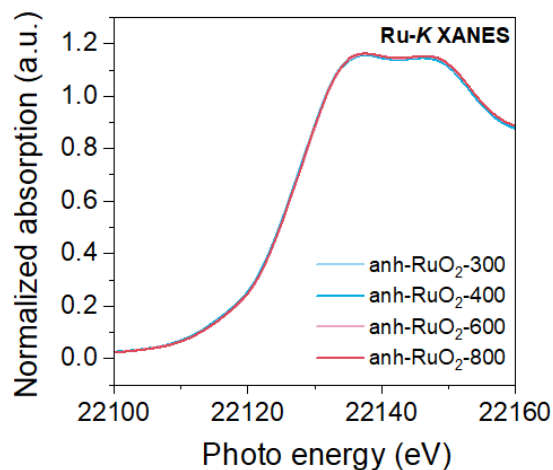


Figure S13. Ru *K*-edge XANES spectra of the anh-RuO₂ after thermal treatments.

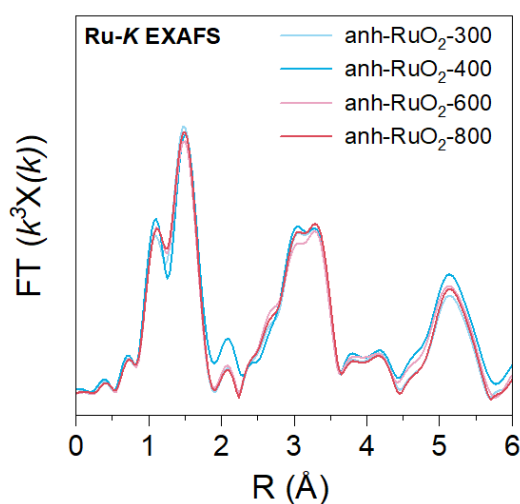


Figure S14. Ru *K*-edge EXAFS spectra of the anh-RuO₂ after thermal treatments.

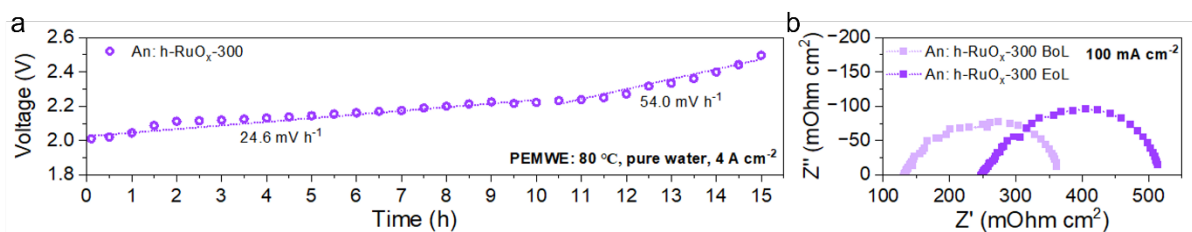


Figure S15. **a**, PEMWE stability test with the h-RuO_x-300 anode for operating at 4 A cm⁻², and **b**, Nyquist plots collected at the BoL and EoL from galvanostatic EIS measurements at 100 mA cm⁻².

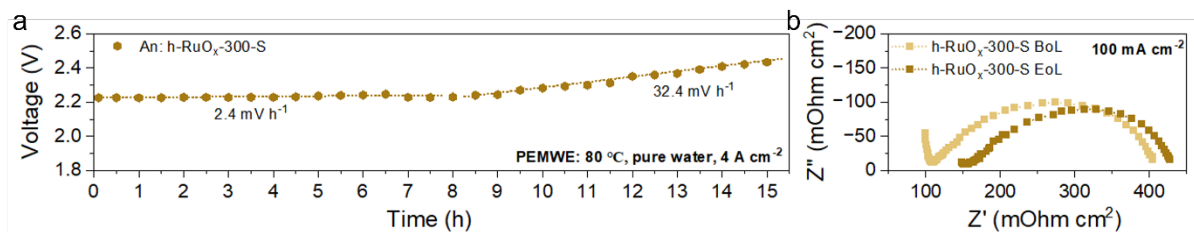


Figure S16. **a**, PEMWE stability test with the SBR stabilized h-RuO_x-300-S anode for operating at 4 A cm⁻², and **b**, Nyquist plots collected at the BoL and EoL from galvanostatic EIS measurements at 100 mA cm⁻².

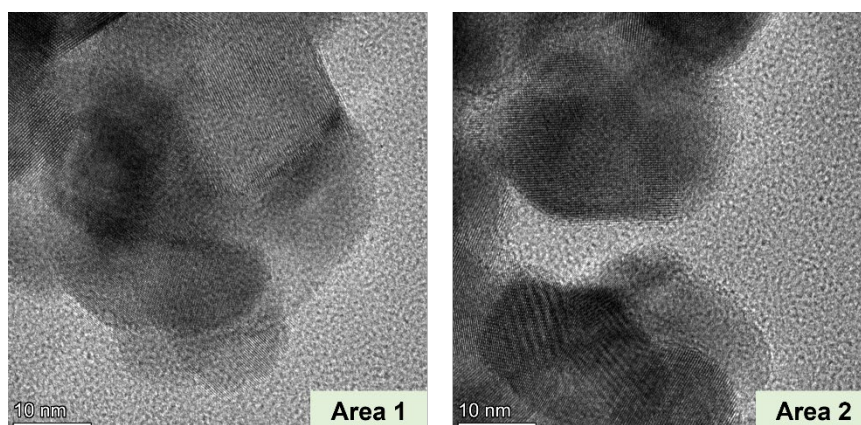


Figure S17. HR-TEM images of the anh-RuO₂ collected from the CCM after going through the stability test in PEMWE at 4 A cm⁻².

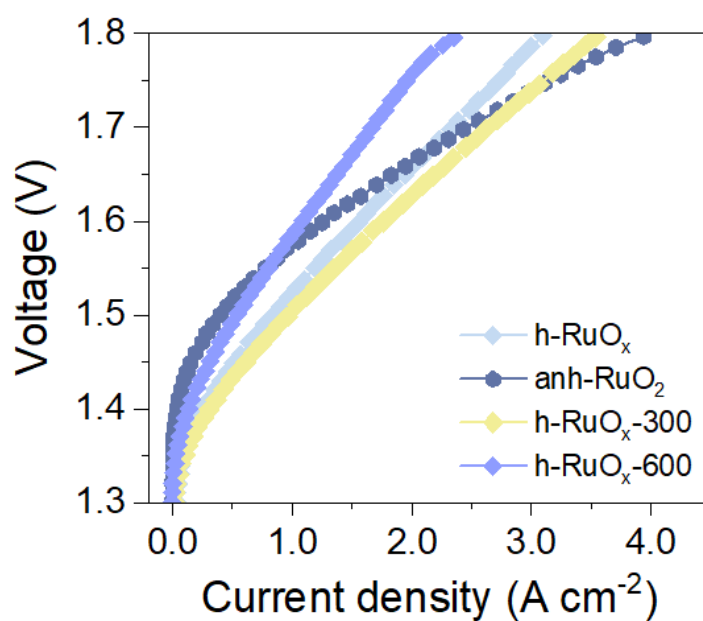


Figure S18. The PEMWE performance curves of the h-RuO_x-300 and h-RuO_x-600 catalysts after the thermal treatments conducted under 300 °C and 600 °C, respectively, compared with

the h-RuO_x anode and anh-RuO₂ anode.

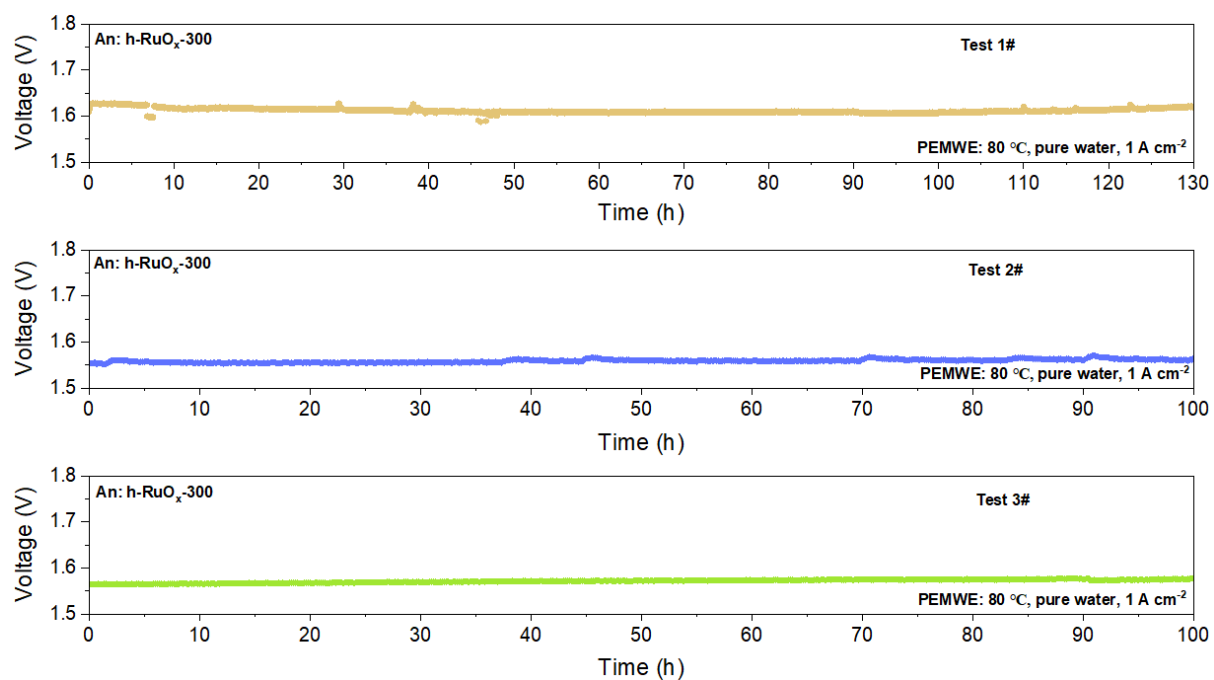


Figure S19. Repeated PEMWE stability testing with the h-RuO_x-300 anode for operating at 1 A cm⁻².

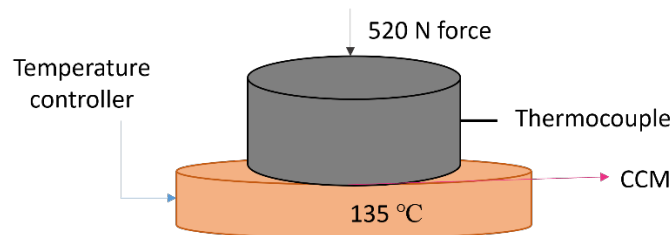


Figure S22. Equipment set up for the hot-pressing procedure to CCM.

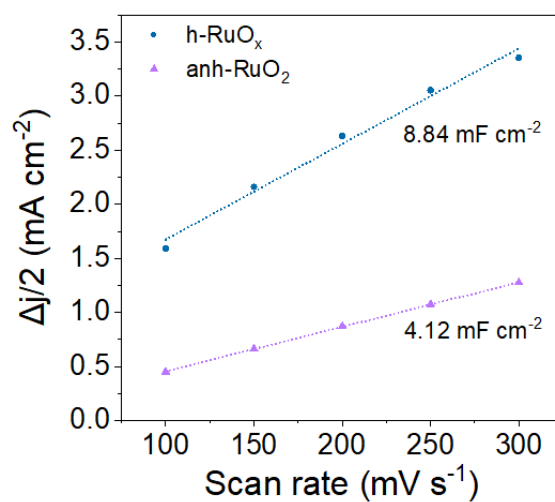


Figure S23. The electrochemical double-layer capacitance obtained from the liner fitting of $\Delta j/2$ against the scan rates.

The ECSA of the Ru oxides was estimated from the electrochemical double-layer capacitance (C_{dl}) measurements under RDE conditions and the specific capacitance (C_s) used for calculation is 0.035 mF cm^{-2} . The as-calculated ECSA for h-RuO_x is $25.2 \text{ m}^2 \text{ g}^{-1}$, and for anh-RuO₂ is $11.7 \text{ m}^2 \text{ g}^{-1}$.

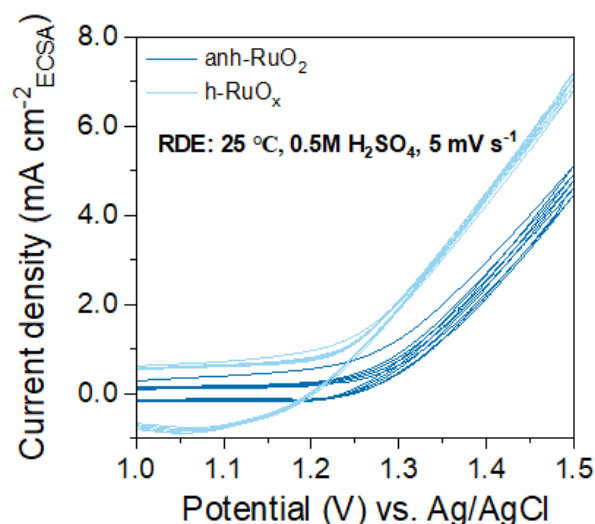


Figure S24. A comparison of the cyclic voltammetry curves from the anh-RuO₂, and h-RuO_x with the current density normalized to ECSA.

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